New Wheels Exploration Into Sql.

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Problem Statement:

Business Context:
A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective:
New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Data Description:
The data provided has

- Attributes on the vehicles New-Wheels sells - What are the make, model, and year? What is the price point?
- Attributes on the customers, such as where they live and payment methods
- Attributes on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

Question 1, Part 1: Find the total number of customers who have placed orders.

Query:

```
SELECT COUNT (DISTINCT customer_id)
FROM customer_t;
```

Output:

Showing 1 rows

COUNT (DISTINCT cu...
994

Observations and Insights:

- There were 994 customers over the past 4 quarters.

Question 1, Part 2: What is the distribution of the customers across states?

Query:

```
SELECT STATE, COUNT(CUSTOMER_ID) AS customer_count
FROM customer_t
GROUP BY STATE
ORDER BY customer_count DESC;
```

Output:

Showing first 10 rows out of 49 rows

state	customer_count
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28
Arizona	26

Observations and Insights:

- Most customers are from more heavily populated states.
- A surprisingly high number of customers are from DC.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Query:

```sql
SELECT
    p.VEHICLE_MAKER,
    COUNT(DISTINCT o.CUSTOMER_ID) AS customer_count
FROM product_t AS p
JOIN order_t AS o ON p.PRODUCT_ID = o.PRODUCT_ID
GROUP BY p.VEHICLE_MAKER
ORDER BY customer_count DESC
LIMIT 5;
```

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolets sell far more successfully.
- There is not too much of a difference between how successful the other most popular car brands are.

Question 3: Which is the most preferred vehicle maker in each state?

Query:

```
CREATE TEMP VIEW TopMakersPerState AS
SELECT
    STATE,
    CASE
        WHEN COUNT(*) > 1 THEN 'multiple'
        ELSE MAX(VEHICLE_MAKER)
    END AS VEHICLE_MAKER,
    MAX(order_count) AS order_count
FROM (
    SELECT
        c.STATE,
        p.VEHICLE_MAKER,
        COUNT(o.order_ID) AS order_count,
        RANK() OVER (PARTITION BY c.STATE ORDER BY
COUNT(c.CUSTOMER_ID) DESC) as mkr_rank
    FROM customer_t AS c
    JOIN order_t AS o ON c.CUSTOMER_ID = o.CUSTOMER_ID
    JOIN product_t AS p ON o.PRODUCT_ID = p.PRODUCT_ID
    GROUP BY c.STATE, p.VEHICLE_MAKER
)
WHERE mkr_rank = 1
GROUP BY STATE;
```

```
SELECT * FROM TopMakersPerState ORDER BY STATE LIMIT 10
OFFSET 0;
SELECT * FROM TopMakersPerState ORDER BY STATE LIMIT 10
OFFSET 10;
SELECT * FROM TopMakersPerState ORDER BY STATE LIMIT 10
OFFSET 20;
SELECT * FROM TopMakersPerState ORDER BY STATE LIMIT 10
OFFSET 30;
SELECT * FROM TopMakersPerState ORDER BY STATE LIMIT 10
OFFSET 40;
```

Output:

Showing first 10 rows out of 143 rows

STATE	VEHICLE_MAKER	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Output:

Showing first 10 rows out of 10 rows

STATE	VEHICLE_MAKER	order_count
Georgia	Toyota	3
Hawaii	multiple	1
Idaho	Dodge	2
Illinois	multiple	3
Indiana	Mazda	4
Iowa	multiple	1
Kansas	multiple	1
Kentucky	multiple	1
Louisiana	multiple	2
Maine	Mercedes-Benz	1

Output:

Showing first 10 rows out of 10 rows

STATE	VEHICLE_MAKER	order_count
Maryland	Ford	5
Massachusetts	multiple	2
Michigan	Ford	3
Minnesota	GMC	3
Mississippi	multiple	1
Missouri	Chevrolet	4
Montana	multiple	1
Nebraska	multiple	1
Nevada	Pontiac	3
New Hampshire	multiple	1

Output:

Showing first 10 rows out of 10 rows

STATE	VEHICLE_MAKER	order_count
New Jersey	multiple	2
New Mexico	Dodge	2
New York	multiple	5
North Carolina	Volvo	3
North Dakota	multiple	1
Ohio	Chevrolet	6
Oklahoma	multiple	2
Oregon	Toyota	2
Pennsylvania	Toyota	3
South Carolina	multiple	1

Output:

Showing 9 rows

STATE	VEHICLE_MAKER	order_count
Tennessee	Mazda	3
Texas	Chevrolet	9
Utah	multiple	1
Vermont	Mazda	1
Virginia	Ford	5
Washington	Chevrolet	5
West Virginia	Mercedes-Benz	2
Wisconsin	multiple	1
Wyoming	Buick	1

Observations and Insights:

- Texas sells Chevrolets frequently.
- Almost half of the states had multiple most common makes of car sold.
- Most states that had multiple makes that had sold the same amount were also states who sold very few of that make of car.

Question 4, Part 1: Find the overall average rating given by the customers.

Query:

```
SELECT
    AVG(rating_value) AS overall_avg_rating
FROM (
    SELECT
        CASE
            WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
            WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 2
            WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 3
            WHEN CUSTOMER_FEEDBACK = 'Good' THEN 4
            WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 5
        END AS rating_value
    FROM order_t
) AS feedback_numeric;
```

Output:

Showing 1 rows

overall_avg_rating
3.135

Observations and Insights:

- The average rating of the services provided was 3.135 out of 5.
- There is a great amount of room for improvement.

Question 4, Part 2: What is the average rating in each quarter?

Query:

```sql
SELECT
    QUARTER_NUMBER,
    AVG(rating_value) AS avg_rating
FROM (
    SELECT
        QUARTER_NUMBER,
        CASE
            WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
            WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 2
            WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 3
            WHEN CUSTOMER_FEEDBACK = 'Good' THEN 4
            WHEN CUSTOMER_FEEDBACK = 'Very Good' THEN 5
        END AS rating_value
    FROM order_t
) AS quarter_feedback
GROUP BY QUARTER_NUMBER
ORDER BY QUARTER_NUMBER;
```

Output:

Showing 4 rows

QUARTER_NUMBER	avg_rating
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

Observations and Insights:

- The average rating of the services provided went lower in the later quarters.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Query:

```
SELECT
    QUARTER_NUMBER,
    COUNT(*) AS total_feedback,
    ROUND(SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Very Bad' THEN
1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_bad_pct,
    ROUND(SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Bad' THEN 1
ELSE 0 END) * 100.0 / COUNT(*), 2) AS bad_pct,
    ROUND(SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Okay' THEN 1
ELSE 0 END) * 100.0 / COUNT(*), 2) AS okay_pct,
    ROUND(SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Good' THEN 1
ELSE 0 END) * 100.0 / COUNT(*), 2) AS good_pct,
    ROUND(SUM(CASE WHEN CUSTOMER_FEEDBACK = 'Very Good'
THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_good_pct
FROM order_t
GROUP BY QUARTER_NUMBER
ORDER BY QUARTER_NUMBER;
```

Output:

Showing 4 rows

quarter_number	total_feedback	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	310	10.97	11.29	19.03	28.71	30
2	262	14.89	14.12	20.23	22.14	28.63
3	229	17.9	22.71	21.83	20.96	16.59
4	199	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- Each of the curves has the general shape if not the form of a normal distribution.
- The average rating has a positive correlation with the total amount of feedback.

Question 6: What is the trend of the number of orders by quarter?

Query:

```
SELECT
    QUARTER_NUMBER,
    COUNT(ORDER_ID) AS total_orders
FROM order_t
GROUP BY QUARTER_NUMBER
ORDER BY QUARTER_NUMBER;
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- The total number of orders made has decreased with each quarter.

Question 7, Part 1: Calculate the net revenue generated by the company.

Query:

```
SELECT
    SUM(o.QUANTITY * (p.VEHICLE_PRICE * (1 - (o.DISCOUNT /
100)))) AS total_net_revenue
FROM order_t o
JOIN product_t p ON o.PRODUCT_ID = p.PRODUCT_ID;
```

Output:

Showing 1 rows

total_net_revenue
124714086.25967374

Observations and Insights:

- The total net revenue generated this year was approximately $124,714,086.

Question 7, Part 2: What is the quarter-over-quarter % change in net revenue?

Query:

```
SELECT
    QUARTER_NUMBER,
    current_revenue,
    previous_revenue,
    ((current_revenue - previous_revenue) /
previous_revenue) * 100 AS qoq_percentage_change
FROM (
    SELECT
        QUARTER_NUMBER,
        current_revenue,
        LAG(current_revenue) OVER (ORDER BY QUARTER_NUMBER)
AS previous_revenue
    FROM (
        SELECT
            o.QUARTER_NUMBER,
            SUM(o.QUANTITY * (p.VEHICLE_PRICE * (1 -
(o.DISCOUNT / 100)))) AS current_revenue
        FROM order_t o
        JOIN product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
        GROUP BY o.QUARTER_NUMBER
    ) AS QuarterlyRevenue
) AS LaggedRevenue
ORDER BY QUARTER_NUMBER;
```

Output:

Showing 4 rows

QUARTER_NUMBER	current_revenue	previous_revenue	qoq_percentage_change
1	39421580.15929598		
2	32715830.392376337	39421580.15929598	-17.010352552644612
3	29229896.19364898	32715830.392376337	-10.655190948598614
4	23346779.51435219	29229896.19364898	-20.127052933479323

Observations and Insights:

- There is no clear trend in how significant the negative percentage change of revenue will be from quarter to quarter.

Question 8: What is the trend of net revenue and orders by quarters?

Query:

```
SELECT
    o.QUARTER_NUMBER,
    -- Count the total number of orders
    COUNT(o.ORDER_ID) AS total_orders,
    -- Calculate Net Revenue: Quantity * (Price * (1 -
Discount)) [cite: 32, 36, 46]
    SUM(o.QUANTITY * (p.VEHICLE_PRICE * (1 - o.DISCOUNT)))
AS net_revenue
FROM
    order_t o
JOIN
    product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
GROUP BY
    o.QUARTER_NUMBER
ORDER BY
    o.QUARTER_NUMBER;
```

Output:

Showing 4 rows

quarter_number	total_orders	net_revenue
1	310	18032549.899600018
2	262	13122995.776373798
3	229	8882298.8449
4	199	8573149.238220006

Observations and Insights:

- A quarter's net revenue has a positive correlation to the total number of orders made in the quarter.

Question 9: What is the average discount offered for different types of credit cards?

Query:

```
SELECT
    curr.QUARTER_NUMBER,
    curr.net_revenue AS current_quarter_revenue,
    prev.net_revenue AS previous_quarter_revenue,
    ((curr.net_revenue - prev.net_revenue) /
prev.net_revenue) * 100 AS qoq_perc_change
FROM
    (
        SELECT
            o.QUARTER_NUMBER,
            SUM(o.QUANTITY * (p.VEHICLE_PRICE * (1 -
o.DISCOUNT))) AS net_revenue
        FROM order_t o
        JOIN product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
        GROUP BY o.QUARTER_NUMBER
    ) AS curr
LEFT JOIN
    (
        SELECT
            o.QUARTER_NUMBER,
            SUM(o.QUANTITY * (p.VEHICLE_PRICE * (1 -
o.DISCOUNT))) AS net_revenue
        FROM order_t o
        JOIN product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
        GROUP BY o.QUARTER_NUMBER
    ) prev ON curr.QUARTER_NUMBER = (prev.QUARTER_NUMBER +
1)
ORDER BY
    curr.QUARTER_NUMBER;
```

Output:

Showing 4 rows

QUARTER_NUMBER	current_quarter_revenue	previous_quarter_reve...	qoq_perc_change
1	18032549.899600018		
2	13122995.776373798	18032549.899600018	-27.226067031901678
3	8882298.8449	13122995.776373798	-32.31500644927895
4	8573149.238220006	8882298.8449	-3.480513458039084

Observations and Insights:

- There is no clear trend in how significant the negative correlation between the percent change in revenue will be from quarter to quarter.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Query:

```
SELECT
    QUARTER_NUMBER,
    AVG(julianday(SHIP_DATE) - julianday(ORDER_DATE)) AS
average_shipping_time
FROM
    order_t
GROUP BY
    QUARTER_NUMBER
ORDER BY
    QUARTER_NUMBER;
```

Output:

Showing 4 rows

quarter_number	average_shipping_time
1	57.16774193548387
2	71.11068702290076
3	117.75545851528385
4	174.09547738693468

Observations and Insights:

- The average shipping time has increased exponentially with each quarter.

Business Metrics Overview:

Total Revenue	Total Orders	Total Customers	Average Rating
$124,714,086	1000	994	3.135

Last Quarter Revenue	Last Quarter Orders	Average Days To Ship	Good Feedback
$23,346,779	199	98	44.1%

Business Recommendations:

- The average shipping time observed during a quarter is negatively correlated to the average rating, and as the shipping time has gone up with each quarter, the amount of revenue and number of orders has gone down; therefore, the average shipping time should be reduced as much as possible.
- In order to achieve success in this domain without compromising the assets that the company is already struggling to expand with, the satisfaction of customers living in states with smaller areas and greater numbers of sales should be given first priority in order to ensure that a loyal company reputation can be established, and the assets given to those initiatives can then be secured.